ANNUAL MAGAZINE

# MARS

DREAM | INNOVATE | BUILD



## Mechatronics and Robotics Society

## 2025-26

This magazine captures our year of challenges, breakthroughs, and the spirit of teamwork that fuels our mission. As we set our sights on the future, we continue to dream, design, and deliver.
Welcome to the world of MARS—where innovation meets exploration.

MARS

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# Outline

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# About MARS



The Mechatronics and Robotics Society (MARS) is the sole robotics society of Thapar Institute of Engineering and Technology, Patiala. MARS is highly dedicated towards robotics and innovation. Our organization actively participates in prestigious robotics-based national and international competitions like ERC, IRC, SIH, DD Robocon and keeps bringing laurels to the institution.

We motivate students of our institution to sharpen their technical skills by providing our resources and a platform to learn, perform, and experiment with individual technical abilities and eager to build a brighter future.

# PRESIDENT's Note



**Prof. Ashish Singla**
Professor, MED

Thapar Institute of Engineering and Technology

Ph.D.
Indian Institute of Technology, Kanpur

As President of the Mechatronics and Robotics Society (MARS), I am honored to witness our student's unwavering commitment to transforming ideas into reality. MARS is not just a platform for technical learning; it is a thriving community where curiosity meets purpose, and knowledge becomes the foundation for innovation.

Our mission extends beyond technical skill-building. At MARS, we inspire students to apply their talents to tackle meaningful challenges from developing sustainable technologies to advancing autonomous systems that enrich human life. Through collaborative projects and hands-on exploration, our members cultivate resilience, creativity, and a sense of responsibility as future engineers.

With the support of our university, MARS has grown into a beacon of progressive thought and dedication, spreading our knowledge and insights to the world, fostering a culture of innovation and inclusivity within robotics. By engaging with pressing issues and sharing their work, our students are creating a lasting impact on the wider community and inspiring a culture that values robotics as a tool for societal advancement. I am immensely proud to support this journey and look forward to the future our young visionaries will build.

# VICE PRESIDENT's Note



**Dr. Sachin Kansal**
Associate Professor, CSED

Thapar Institute of Engineering and Technology

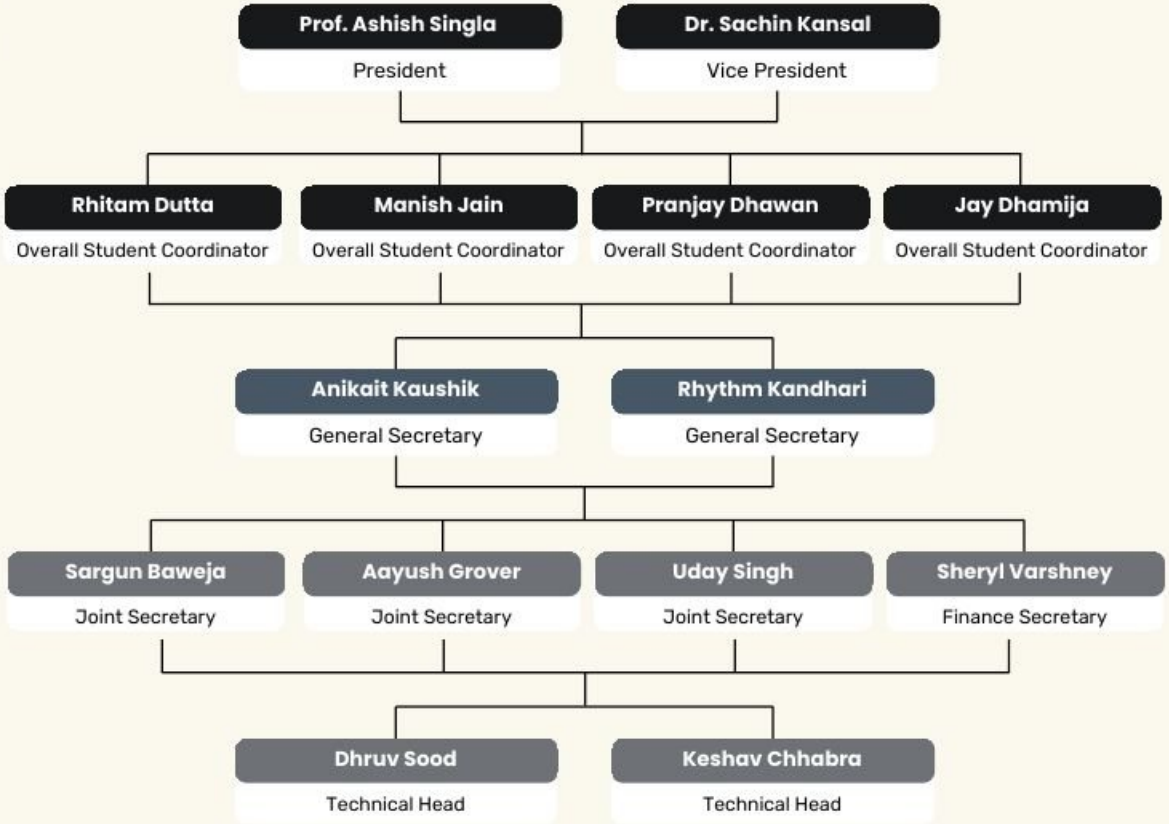
Ph.D.
Indian Institute of Technology, Delhi

As Vice President of the Mechatronics and Robotics Society (MARS), I am continually inspired by the ambition and creativity of our members. MARS is a space where students blend knowledge with purpose, translating theories into real-world applications that push the boundaries of possibility.

Our society takes on challenges that matter from designing autonomous systems to pioneering solutions for sustainability. We actively participate in conferences and publish research, ensuring we remain connected to the latest global innovations. Engaging with the wider scientific community allows us to implement cutting-edge developments into our work, even in the smallest details, as we share our progress with the world.

At MARS, learning extends beyond classrooms; it thrives through exploration, resilience, and practical experience. We are preparing young engineers to lead with empathy, innovation, and a commitment to progress. I am grateful to be part of this journey, supporting our students as they create technologies that make a lasting impact.

# TEAM Structure



MARS

# FY 2025-26

| Sr No | Category | Event/Project Name | Year | Budget (Rs.) | Expenditure (Rs.) |
|---|---|---|---|---|---|
| 1. | Competitions | IRC, ISDC 2026 | 2026 | 6,65,500 | 6,65,500<br>1,00,000* |
| 2. | Projects | Technical Projects | 2025 | 60,000 | 60,000 |
| 3. | Events | Society Fair | 2025 | 5,000 | 5,000 |
| | | Orientation | 2025 | 5,000 | 5,000 |
| 4. | Miscellaneous | Registration IRC ISDC 2026 | 2026 | 64,500 | 64,500 |
| | | **TOTAL** | | **8,00,000** | **9,00,000** |

*Extra Funds over and above budget for Components

BUDGET

# BUDGET

## FY 2024-25

| Sr No | Category | Event/Project Name | Year | Budget (Rs.) | Expenditure (Rs.) |
|---|---|---|---|---|---|
| 1. | Competitions | IRC-2025 (At BITS Goa) | 2025 | 2,35,000 | 2,35,000<br>1,25,000*<br>1,00,000** |
| 2. | Projects | Robomuse | 2024 | 30,000 | 29,000 |
| 3. | Events | Society Fair | 2024 | 5,000 | 5,000 |
| | | Orientation | 2024 | 5,000 | 5,000 |
| 4. | Miscellaneous | Tools | 2024 | 25,000 | 25,000 |
| | | **TOTAL** | | **3,00,000** | **5,24,000** |

*Extra Funds over and above budget for Components
**Extra Funds over and above budget for Travel Purposes

# Hall of FAME




IN THE 2025 EDITION OF THE INTERNATIONAL ROVER CHALLENGE (IRC), TEAM MARS ACHIEVED A REMARKABLE MILESTONE BY SECURING 2ND PLACE AMONG 30 TOP-TIER TEAMS FROM ACROSS THE GLOBE, INCLUDING PRESTIGIOUS INSTITUTIONS LIKE IITS, NITS, AND MOSCOW STATE UNIVERSITY, RUSSIA. DEMONSTRATING SIGNIFICANT GROWTH FROM OUR 18TH POSITION IN THE PREVIOUS YEAR, WE SHOWCASED TREMENDOUS LEAP IN OUR PERFORMANCE.



IN THE FINAL PHASE, WE DELIVERED OUTSTANDING PERFORMANCES ACROSS ALL FIVE CRITICAL TASKS RDO, ABEX, IDMO, AUTEX AND PIMA SHOWCASING OUR TECHNICAL EXPERTISE, INNOVATIVE PROBLEM-SOLVING, AND ON-GROUND EFFICIENCY. THIS SUCCESS HIGHLIGHTS OUR TEAM'S RESILIENCE, DEDICATION, AND CONTINUOUS PURSUIT OF EXCELLENCE, FIRMLY ESTABLISHING TEAM MARS AS ONE OF THE LEADING STUDENT ROVER TEAMS ON THE GLOBAL STAGE. IT HAS FURTHER INSPIRED US TO SET EVEN GREATER BENCHMARKS FOR FUTURE COMPETITIONS.



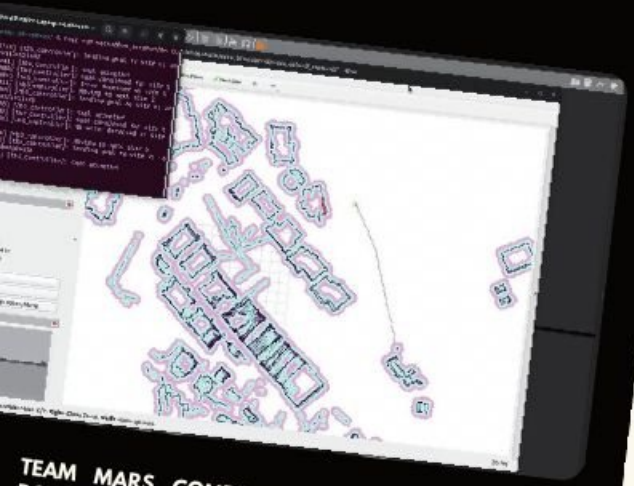


AT THE 2023 EDITION OF SATURNALIA, THE PRESTIGIOUS ANNUAL TECHNO-CULTURAL FEST HELD AT THE THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, TEAM MARS ACHIEVED A REMARKABLE FEAT BY EMERGING VICTORIOUS IN THE HIGHLY COMPETITIVE ROBOWARS COMPETITION.




TEAM MARS WON THE 'EMERGING TEAM OF THE YEAR' IN THE INTERNATIONAL ROVER CHALLENGE 2023, SHOWCASING INNOVATION AND TECHNICAL EXPERTISE THAT ESTABLISHED US AS A RISING FORCE IN ROBOTICS AND SPACE EXPLORATION.





TEAM MARS COMPETED IN ERC, A PRESTIGIOUS ROBOTICS HACKATHON ORGANIZED BY BITS GOA DURING THE SUMMER OF 2024. WE BAGGED THE FIRST POSITION ALL OVER INDIA. PARTICIPANTS INCLUDED TOP COLLEGES FROM ACROSS INDIA, SUCH AS IIT 'S AND NIT'S.

MARS



TEAM MARS RECENTLY CELEBRATED A SIGNIFICANT ACHIEVEMENT BY BEING SELECTED AS ONE OF THE TOP 25 TEAMS WORLDWIDE TO COMPETE IN THE INTERNATIONAL ROVER CHALLENGE (IRC) 2024, HELD IN COIMBATORE. WE DEMONSTRATED OUR EXPERTISE THROUGHOUT THE COMPETITION THROUGH INNOVATIVE SOLUTIONS AND PRECISION ENGINEERING. WE DEMONSTRATED OUR EXPERTISE THROUGHOUT THE COMPETITION THROUGH INNOVATIVE SOLUTIONS AND PRECISION ENGINEERING.

MARS

# International Rover Challenge 2026 Winners











Manipal Institute of Technology, Manipal, Udupi, India

Jan 28 - Feb 02, 2026

At the International Rover Challenge 2026, months of design, testing, and iteration were brought to life on the field. Competing alongside IITs, NITs, and international teams like the University of Warsaw, we rose to the challenge and emerged as champions.

The competition featured four demanding tasks that evaluated our technical systems along with critical skills like presentations and strategy, making it a complete test of engineering and teamwork.



















Team MARS made an impressive debut at the International Space Drone Challenge 2026 by securing the 1st Runner Up position. Competing against top teams from IITs, NITs, and renowned global institutions, we held our ground in a highly competitive environment.

The competition featured two major mission tasks that tested navigation, autonomy, and precision. Our drone's strong design and teamwork, enabled us to achieve a podium finish.

# International Space Drone Challenge 2026 Runners-Up



Manipal Institute of Technology, Manipal, Udupi, India

Jan 31 - Feb 02, 2026

# Space Robotics Society

## International Rover Challenge (IRC):

It challenges university students to conceptualize, design, develop and operate an astronaut-assistive next-generation planetary rover and perform specific missions in Mars simulated environmets. Team MARS of TIET has been participating in this competition since the last three years. The objective of the competition is to provide students with a real-world interdisciplinary space engineering experience, combining practical engineering skills with soft skills, including business planning and project management.

## International Space Drone Challenge (ISDC):

It challenges university students to design, build, and operate astronaut-assistive space drones through mission-based tasks conducted in a simulated Martian environment. By integrating aerospace systems, electronics, control, communication, and mission planning, ISDC provides a hands-on, real-world platform that fosters innovation, technical excellence, and teamwork, preparing students for future challenges in space robotics.







# PROJECTS



ROVER MANDAKINI : THE 2023 ITERATION OF OUR ROVER TO COMPETE IN THE INTERNATIONAL ROVER CHALLENGE





ROVER BHAGIRATHI : THE BETTER AND IMPROVED 2024 ITERATION OF OUR ROVER TO COMPETE IN THE IRC'24




MARVIN IS AN AUTONOMOUS CLEANING ROBOT EQUIPPED WITH ULTRASONIC SENSORS FOR PRECISE OBSTACLE DETECTION AND AVOIDANCE THROUGH A FEEDBACK-BASED SYSTEM. IT BOASTS EFFICIENT CLEANING CAPABILITIES POWERED BY A HIGH-RPM BRUSHLESS DC MOTOR.



ROBOMUSE IS AN ADVANCED NAVIGATION BOT POWERED BY ROS.

IT EXCELS IN COMPLEX ENVIRONMENTS WITH SMART ALGORITHMS, REAL TIME OBSTACLE AVOIDANCE, 2D MAPPING AND EFFICIENT PATH PLANNING WITH THE HELP OF A LIDAR.

IT HAS INTEL NUC, MAGNETIC ENCODERS, IMU AND TEENSY 4.1 FOR POWERING THE ROBOT'S COMPUTING AND CONTROL SYSTEMS.

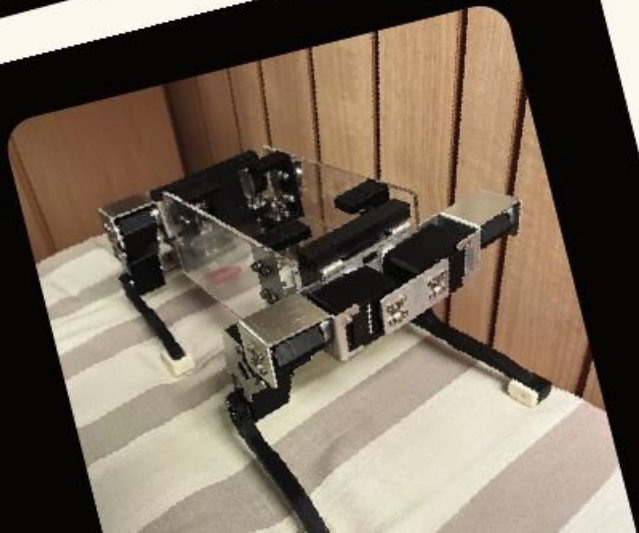


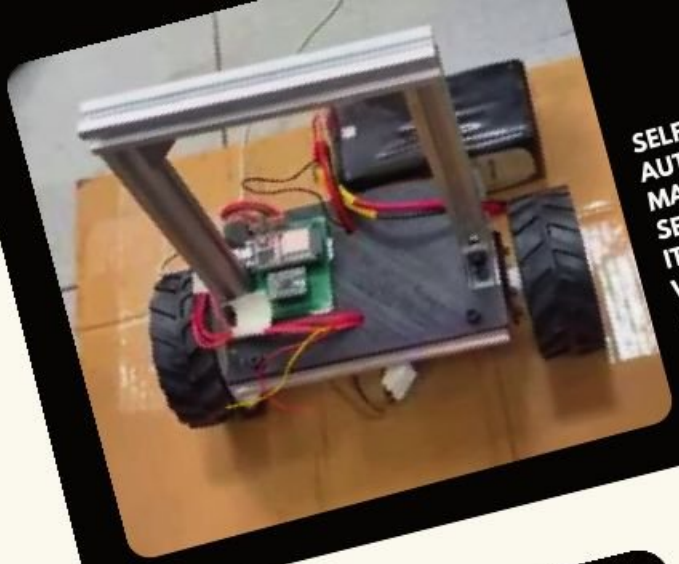
SELF-STABLIZING GIMBAL



RE-CONFIGURABLE ROBOT

THE QUADRUPED ROBOT HAS FOUR LEGS FOR PASSIVE STABILITY IN A STANDING POSITION. IT USES 12 SERVO MOTORS AND 3 DEGREES OF FREEDOM IN EACH LEG ASSEMBLY TO CREATE DIFFERENT MOVEMENTS AND MAINTAIN STABILITY.





SELF-BALANCING ROBOT IS AN AUTONOMOUS SYSTEM DESIGNED TO MAINTAIN ITS UPRIGHT POSITION USING SENSORS AND CONTROL ALGORITHMS. IT CONTINUOUSLY MEASURES ITS TILT WITH SENSORS SUCH AS IMU AND ADJUSTS MOTOR SPEEDS IN REAL TIME TO STAY BALANCED, USING CONTROL SYSTEMS, INTRO FOR NEW BIES.

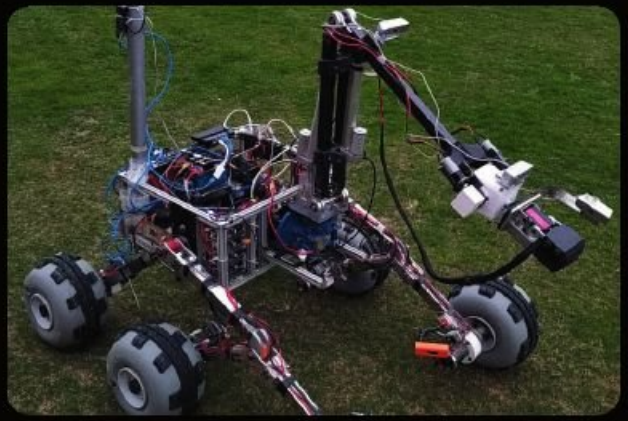


ROVER ALAKANANDA: THE THIRD ITERATION OF OUR ROVER FOR IRC'25.

A ROBOTIC MANIPULATOR IS A MECHANICAL SYSTEM WITH 3 DOF DESIGNED TO PERFORM PRECISE MOVEMENTS FOR TASKS SUCH AS LIFTING, PLACING, AND ASSEMBLING OBJECTS. IT CONSISTS OF MULTIPLE JOINTS AND LINKS CONTROLLED BY MOTORS AND PROGRAMMED ALGORITHMS TO ACHIEVE ACCURATE POSITIONING. THE MOTIVE WAS TO INTRODUCE ROBOTICS TO NEW-BIES.



# Rover GANGA





Ganga, our fourth iteration of the Mars rover prototype, features several upgrades.

- Equipped with a 5-DOF lightweight robotic arm crafted from carbon fiber.
- Soil analysis system, where an auger-based mechanism enables precise sampling.
- Horizontal science module processes samples with reagents to detect signs of life.
- ROS 2 framework based control system.
- Electronics utilize custom double-sided fabricated PCBs.
- Commercially available beach wheels .
- In-house fabricated treads for enhanced grip.
- Custom-made mild steel wheel flanges.

# Drone GARUD





Garud, our first iteration of the space drone prototype, showcases several features.

- Carbon-fiber quadcopter frame with 3D-printed PLA hinges for improved thrust-to-weight efficiency.
- Performs precision payload drops using a lightweight servo-controlled jaw mechanism.
- Uses dual FPV cameras on separate channels for real-time multi-angle visual monitoring.
- Powered by Pixhawk with ArduPilot EKF for stable and precise flight control.
- Collects and logs atmospheric data onboard using a custom sensor PCB.
- Provides live ground-station monitoring of navigation and sensors via telemetry.

# SOCIAL MEDIA OUTREACH




https://marstiet.com ⋮ 10k+ view

mars-tu

Mechatronics and Robotics Society, TIET

mars_tiet ⋮ 2.3k+ followers, 600k+ views

mars@thapar.edu

**Scan to know more!**

# Mars ALUMNI



**Gaurav Sivaych**

PeerROBOTICS

Founder, MARS
Class of 2020

**He was the Founder of MARS and has been working with Peer Robotics as a Sr. Electrical Systems Engineer since the past 5 years.**



**Manjot Singh Suri**

**ADDVERB**

General Secretary, MARS
Class of 2023

**He was a part of the control and automation department and is now working at Addverb fulltime.**



**Rajat Arora**

**ADDVERB**

Technical Lead, MARS
Class of 2023

**He used to work in the control and automation department of mars and is now working at Addverb full time.**

# Mars ALUMNI



**Mukund Gupta**

Joint Secretary, MARS
Class of 2023

**He was associated with the electronics department and is now working with Maruti Suzuki India Limited.**



**Manveer Singh**

Joint Secretary, MARS
Class of 2024

**He worked with addverb as a Mobile Robotics intern and is now pursuing masters from University of Michigan Robotics Department.**



**Pradyun Sharma**

TU Delft

Mechanical Lead, MARS
Class of 2024

**He interned at IIT Delhi , Co-founded an Ed Tech robotics startup and is now pursuing MSc at TU Delft, Netherlands.**

# Mars ALUMNI



**Anant Tomar**

**ADDVERB**

Rover Lead, MARS
Class of 2025

**He was the rover lead and was responsible for the rocker-bogie design, analysis and modelling and is now interning with Addverb.**



**Maadhav Katyal**

iNTUTE.ai

Joint Secretary, MARS
Class of 2025

**With over 6 years of hands-on experience in Electronics and Robotics, he was the Team Captain in IRC and is now interning with Intute.ai**



**Rhitam Dutta**

The University of Queensland

Joint Secretary, MARS
Class of 2026

**He was the person who built the autonomous control systems, and computational optimization, while also managing corporate relations and external affairs.**

# Mars ALUMNI



**Jay Dhamija**

NOVUS HI-TECH

General Secretary, MARS
Class of 2026

**Works at the intersection of robotics, automation, and research, with proficiency in ROS2, MATLAB, embedded systems, and control systems.**



**Manish Jain**

NOVUS HI-TECH

General Secretary, MARS
Class of 2026

**Proficient in CAD, FEA, 3D Printing and Manufacturing, with strong expertise in managing operations, finances, and workflow.**



**Pranjay Dhawan**

ZIPBOLT

Technical Head, MARS
Class of 2026

**Oversees the electronics department with proficiency in PCB design hardware electronics and component fabrication.**

# MARS Patents

**Our research papers and patents reflect the collaborative efforts of MARS members in addressing challenges within robotics and automation. Presented at various international conferences, these works contribute to advancing technical knowledge and practical solutions in the field.**

### Indian Patent Office

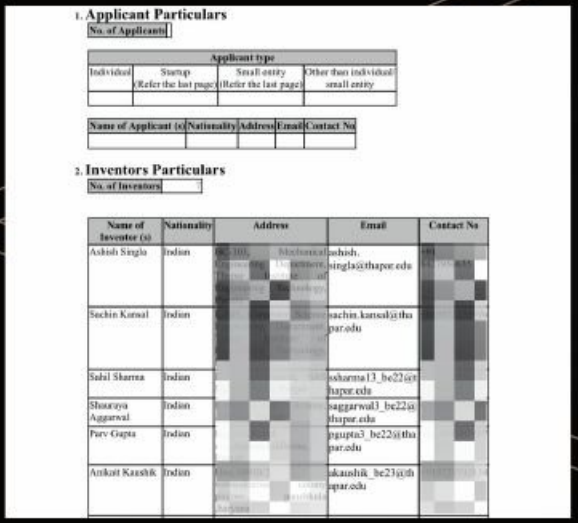


Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India

| Application Details | |
|---|---|
| APPLICATION N | |
| APPLICATION TYPE | ORDINARY APPLICATION |
| DATE OF FILING | 14/05/2024 |
| APPLICANT NAME | Thapar Institute of Engineering and Technology |
| TITLE OF INVENTION | AN ACTIVE EXOSKELETON SYSTEM FOR THE REHABILITATION O |
| FIELD OF INVENTION | MECHANICAL ENGINEERING |
| E-MAIL (As Per Record) | ishasharmasharma1987@gmail.com |
| ADDITIONAL-EMAIL (As Per Record) | |
| E-MAIL (UPDATED Online) | |
| PRIORITY DATE | |
| REQUEST FOR EXAMINATION DATE | 10/07/2024 |
| PUBLICATION DATE (U/S 11A) | 31/05/2024 |

An active exoskeleton system for the rehabilitation of the amputees

### Indian Patent office



1. **Applicant Particulars**

No. of Applicants

| Applicant type | | | |
|---|---|---|---|
| Individual | Startup (Refer the last page) | Small entity (Refer the last page) | Other than individual/ small entity |
| | | | |

| Name of Applicant (s) | Nationality | Address | Email | Contact No |
|---|---|---|---|---|
| | | | | |

2. **Inventors Particulars**

No. of Inventors

| Name of Inventor (s) | Nationality | Address | Email | Contact No |
|---|---|---|---|---|
| Ashish Singla | Indian | [redacted] | ashish.singla@thapar.edu | [redacted] |
| Sachin Kansal | Indian | [redacted] | sachin.kansal@thapar.edu | [redacted] |
| Sahil Sharma | Indian | [redacted] | [redacted] | [redacted] |
| Shaurya Aggarwal | Indian | [redacted] | [redacted] | [redacted] |
| Parv Gupta | Indian | [redacted] | [redacted] | [redacted] |
| Ankit Kaushik | Indian | [redacted] | [redacted] | [redacted] |

On-Board Rover Solid Soil Analysis: Chemical Detection & Sensor Monitoring

# MARS Publications



Design and Analysis of a Mars Rover with Rocker-Bogie Suspension carrying a 5-DOF Robotic Arm

Design and Analysis of a Mars Rover with Rocker-Bogie suspension carrying a 5-DOF Robotic Arm



Long-Range Real-Time ROS2 Teleoperation Framework for Rovers in Harsh Terrains

Long -Range Real-Time ROS2 Teleoperation Framework for Rovers in Harsh Terrains



Designed and Field-Tested Electrical Architecture for a Multi-Subsystem Rover

Design and Field-tested Architecture for Electrical Architecture for a Multi subsystem Rover



ROS2 Framework Based Design and Evaluation of an Autonomous Robotic System for Real-Time Grid Failure Assesment

ROS2 Framework Based Design and Evaluation of an Autonomous Robotic System for Real-Time Grid Failure Assesment



Design and Implementation of a Power Electronic System for an Autonomous Martian Rover

Design and Implementation of a power Electronic System for an Autonomous Martian Rover

# Team MARS at AIR 2025







**Advances In Robotics (AIR) 2025, IIT Jodhpur**

**2nd - 5th July, 2025**

Thapar Institute of Engineering and Technology made a strong and confident mark at **Advances in Robotics,** India's premier robotics conference. With five research papers accepted, authored exclusively by members of Team MARS, it stood out as the largest undergraduate-only research group at the event. Under the mentorship of Prof. Ashish Singla and Dr. Sachin Kansal, the participation reflected the growing depth and seriousness of undergraduate research at Thapar. The conference brought together leading robotics companies, top professors, and researchers from across the country, creating an intense environment of ideas, innovation, and exchange. This experience was truly distinctive, reinforcing a robust research culture among undergraduate students at Thapar and serving as a powerful ground for academic growth, collaboration, and meaningful professional networking.

# EVENTS Current Year

YEAR 2025-26

## • Felicitation by Director, TIET

**Team MARS was felicitated by Dr. Padmakumar Nair, Director and Dr. Meenakshi Rana, Dean of Student Affairs, TIET, for securing 1st position globally at International Rover Challenge and 2nd position at the International Space Drone Challenge 2026, held at Manipal Institute of Technology, Manipal, Udupi**



**14th May, 2026**

Thapar Institute of Engineering and Technology

## • MARS X HELIX

**17th April, 2026**

Thapar Institute of Engineering and Technology

**We conducted an interactive Rover Showcase and Project Exhibition Stall at Helix, the official technical fest of TIET, engaging participants with live demonstrations of innovative engineering projects and robotic systems.**



# EVENTS Current Year

YEAR 2025-26

## • International Academic Interaction

**Team MARS showcased their rover to the professors from Hiroshima University, Japan, who visited MARS Lab, highlighting its key features and innovations. Alongside our faculty advisors, we discussed our projects, team vision, and future growth in robotics, fostering potential collaborations and knowledge exchange.**



**3rd December, 2025**

Thapar Institute of Engineering and Technology

## • MARS X PWS

**21st November, 2025**

Thapar Institute of Engineering and Technology

**We showcased our Martian rover prototype at the SustainX Hackathon organized by Paryavaran Society at TIET, covering every subsystem from locomotion to control providing hands-on insights into reconfigurable robotics and real-world applications.**



# EVENTS Current Year

YEAR 2025-26

## • WOL3D Guest Lecture

A guest lecture on 3D Printing was successfully conducted by Vivek Mahida, Senior Sales Manager of WOL3D, offering students valuable insights into modern additive manufacturing technologies. The session covered real-world applications, current industry trends, and future opportunities in 3D printing.



**28th August , 2025**

Thapar Institute of Engineering and Technology

## • IHFC, IIT Delhi Visit

**12th August, 2025**

Thapar Institute of Engineering and Technology

Team MARS showcased their rover to the I-Hub Foundation for Cobotics, IIT Delhi, highlighting its key features and innovations. Alongside our faculty advisors, we discussed our projects, team vision, and future growth in robotics, fostering potential collaborations and knowledge exchange.



# EVENTS Previous Year

YEAR 2024-25

## • ANRF Workshop on Practicing Robotics

**At IIT Ropar, we conducted a workshop on Practicing Robotics and Prototype Development, organized by the Robotics Society India and ANRF. We showcased our Martian rover prototype, covering every subsystem from locomotion to control providing hands-on insights into reconfigurable robotics and real-world applications.**



**21st March, 2025**

Indian Institute of Technology(IIT)

Ropar, Punjab

## • Felicitation by DIRECTOR, TIET

**14th February, 2025**

Thapar Institute of Engineering and Technology

**Team MARS was felicitated by Dr. Padmakumar Nair, Director and Dr. Inderveer Channa, Dean of Student Affairs, TIET, for our incredible performance and securing 2nd position globally at International Rover Challenge 2025, held at BITS Pilani K K Birla Goa Campus.**



# EVENTS Previous Year

YEAR 2024-25

## • Session on ACADEMIC RESEARCH work

**An insightful session was conducted with the members of team MARS by Dr. Ashish Singla (President, MARS), emphasizing the significance of research work in the industry. The session also covered various prestigious conferences held globally each year, providing valuable insights into opportunities for knowledge exchange and professional growth.**



**12th February, 2025**

Thapar Institute of Engineering and Technology

## • IHFC, IIT Delhi Visit

**12th February, 2025**

Thapar Institute of Engineering and Technology

**Team MARS showcased their rover to the I-Hub Foundation for Cobotics, IIT Delhi, highlighting its key features and innovations. Alongside our faculty advisors, we discussed our projects, team vision, and future growth in robotics, fostering potential collaborations and knowledge exchange.**



# EVENTS Previous Year

YEAR 2024-25

## • International ROVER Challenge 2025

**In the 2025 International Rover Challenge, Team MARS secured 2nd place among 30 elite teams worldwide, including IITs, NITs, and MSU Russia. This marks a significant achievement, reflecting our remarkable progress from 18th place last year and showcasing our dedication, innovation, and technical excellence.**



**2nd February, 2025**

BITS Pilani K K Birla, Goa Campus

2nd Position Globally

## • Saturnalia TECH Exhibition

**Team MARS was invited to participate in the Tech Exhibition at Saturnalia 2024, the flagship techno-cultural fest of TIET. We had the opportunity to showcase our major projects and engage with a diverse audience, including students and enthusiasts from various colleges, sharing our knowledge and fostering technical discussions.**



**16th November, 2024**

Thapar Institute of Engineering and Technology

# EVENTS Previous Year

YEAR 2024-25

## • Co-founder, Addverb technologies

Sangeet Kumar, Co-founder of Addverb technologies, visited TIET, exploring the MARS Lab and engaging with the entire MARS team. He shared his insights on robotics and automation along with future collaborations with our mentors.



**21st October, 2024**

Thapar Institute of Engineering and Technology

## • Annual ORIENTATION

**11th September, 2024**

Thapar Institute of Engineering and Technology

During the MARS orientation, we inspired students with our mission, vision, departmental structure, workflow, and accomplishments. We highlighted impactful projects, notable achievements, and the support of our esteemed sponsors.



# EVENTS Previous Year

YEAR 2024-25

## • Society FAIR 2024

At the Society Fair for the students of batch of 2028, MARS captivated freshmen with insights into our society's mission, ongoing projects, and learning pathways. Our team of seven guided them in joining MARS to innovate, build skills, and pursue impactful research.



**1st September, 2024**

Thapar Institute of Engineering and Technology

## • ERC Hackathon

**10th August, 2024**

BITS Pilani
KK Birla Goa Campus

**1st Position** All Over India

Team Mars competed in ERC, a prestigious robotics hackathon organized by BITS Goa during the summer of 2024 wherein the team designed and simulated an autonomous grid failure robot capable of maneuvering on difficult terrain and fixing electrical failures in a building.



# MARS Jersey

**Team MARS IRC and ISDC jerseys serve as a common uniform across competitions, expressing a composed identity shaped by clarity, intent, and responsibility. They reflect consistency in approach, seriousness in execution, and a shared understanding of what the team represents.**




**Team MARS IRC and ISDC team Jersey's**
Designed in **white and navy** to reflect clarity and discipline, anchored by team insignia and national identity, uniting rover and drone divisions under a single, competition-driven purpose




**Team MARS IRC and ISDC team Jersey's**
crafted in **grey and navy** to represent balance, endurance, and continuity, serving as a shared identity beyond the field through long build hours, testing phases, and team collaboration.

# Key Moments



At the FROSH Society Fair '24





With Vice Chacellor, TIET for IRC & ISDC 2026



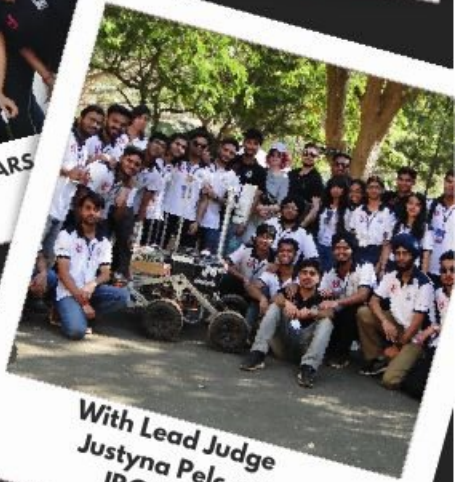
With Mr. Sandeep Kumar, CEO, ADDVERB



Team MARS at IIT Ropar



With the team at MARS Society Stall



With Lead Judge Justyna Pelc, at IRC 2025






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**IRC' 26 Winners**
**ISDC' 26 Runners' Up**





Anikait Kaushik
B.E Mechatronics '27



Rhythm Kandhari
B.E COBS '27

**MARS is dedicated to creating an environment where creativity and technical expertise thrive, allowing each member to grow through meaningful projects based on the latest industrial technologies and requirements, and promoting research and development in the field of robotics. We're more than a team, we're a community of innovators pushing the boundaries of what's possible.**

**At MARS, teamwork and exploration drive us. Through competitions and projects, we grow as engineers and leaders, facing challenges that teach invaluable lessons. We're proud to build both robots and futures through hands-on experience and dedication. We're a family united by a shared passion for technology and a commitment to making a positive impact on the future**

MARS

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